

Poster: Reference-Based Prompting for Acoustic Anomaly Detection

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ABSTRACT

We present a training-free reference-based LLM approach for low-shot acoustic anomaly detection. Each clip is represented by a DSP summary, and the LLM compares a query against a small set of matched normal references through aggregated statistics and explicit query-to-baseline comparison values. This formulation uses the LLM as a structured comparator over lightweight acoustic descriptors and suggests that acoustic anomaly detection is better framed as reference-based comparison than as zero-shot judgment.

CCS CONCEPTS

• **Computing methodologies** → **Anomaly detection**; **Natural language processing**; • **Human-centered computing** → **Ubiquitous and mobile computing systems and tools**.

KEYWORDS

Large Language Models, Acoustic Sensing, Anomaly Detection

ACM Reference Format:

Ruitao Xue, Devki Nandan Jha, Rajiv Ranjan, Tomasz Szydło. 2026. Poster: Reference-Based Prompting for Acoustic Anomaly Detection. In *The 24th Annual International Conference on Mobile Systems, Applications and Services (MobiSys Companion '26)*, June 21–25, 2026, Cambridge, United Kingdom. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3812835.3814869>

1 INTRODUCTION

Acoustic anomaly detection is important for machine condition monitoring and has been studied through benchmarks such as Toy-ADMOs, MIMII, and DCASE Task 2 [1–3]. Most existing approaches rely on task-specific acoustic models or training on normal data [4]. Recent LLM-based sensing work, such as Wi-Chat [5], motivates the question of whether prompting can also support acoustic anomaly detection.

We explore this question under a low-shot, reference-based setting. Instead of feeding raw audio to an LLM, we convert each clip into a compact DSP summary and ask the model to compare a query against a small set of matched normal references. Our goal is not to claim a final state-of-the-art detector, but to study when reference-based prompting is useful relative to a classical one-class baseline. The main finding is that LLM prompting is most useful in a narrow low-shot regime: it can be competitive when only a few matched normal references are available, but classical one-class

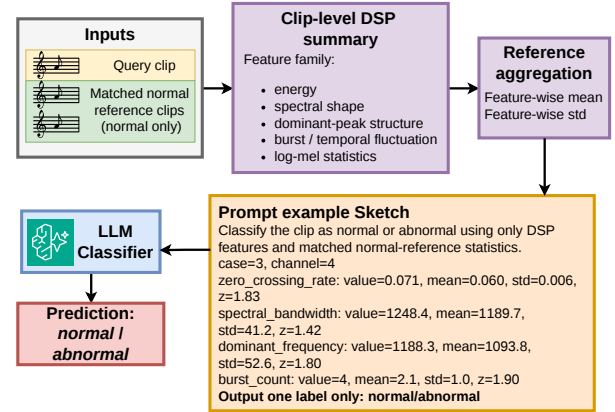


Figure 1: Training-free reference-based prompting pipeline for low-shot acoustic anomaly detection.

models become stronger once more normal data are available for fitting.

2 METHOD OVERVIEW

2.1 Problem Formulation

Given a query clip and a small set of matched normal reference clips, the goal is to predict whether the query is normal or anomalous. Here, *matched* means that the references come from the same machine case and recording channel as the query, and all references are normal-only. We therefore treat acoustic anomaly detection as a reference-based comparison rather than an absolute judgment from a single clip.

This formulation differs from standard supervised classification because no anomalous examples are required in the main setting, and no model parameters are updated for the LLM. It also differs from classical normal-only training because the LLM receives only a small reference set at inference time. This setting is intended to reflect early deployment, where a few normal clips may be easy to collect, but labelled faults are unavailable.

2.2 Reference-Based Prompting System

Our system is training-free and operates on clip-level acoustic summaries rather than raw audio. For each case/channel group, we construct a fixed baseline reference set from a separate pool of normal clips, using deterministic, approximately uniform selection. All query clips in the same group share the same baseline reference set. Reference sizes range from 0-shot to 16-shot.



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ACM ISBN 979-8-4007-2711-5/2026/08.
<https://doi.org/10.1145/3812835.3814869>

Table 1: Held-out test performance.

Method	Shot	Acc.	F1
GPT-5.4-mini	4	0.846	0.850
Gemini 3.1 Flash-Lite	4	0.791	0.812
one-class SVM	4	0.783	0.779
GPT-5.4-mini	16	0.865	0.877
Gemini 3.1 Flash-Lite	16	0.881	0.888
one-class SVM	16	0.940	0.938

Each clip is converted into an 11-dimensional scalar feature vector. These features cover energy, spectral shape, dominant-peak structure, temporal fluctuation, and log-mel summary statistics. For the reference set, we aggregate each feature into a baseline profile using its mean and standard deviation. We then compare the query clip against this baseline by constructing explicit comparison values such as raw query features, feature-wise deltas, and z-scores. These quantities are formatted into a structured prompt with a fixed feature order. The LLM is asked to output a single label only: NORMAL or ABNORMAL.

This design is motivated by three development-stage observations. First, pure zero-shot prompting was unstable because the model tended to over-call or under-call abnormalities without a local normal reference. Second, presenting aggregated baseline statistics worked better than listing multiple raw reference examples directly, suggesting that the organisation of reference information matters. Third, feature usefulness depended on the shot regime, so the strongest-performing feature subset was selected separately for each shot setting using leave-one-feature-out ablation.

3 EVALUATION

3.1 Evaluation Setup

We evaluate the proposed approach on the ToyADMOS ToyConveyor subset [2], using IND recordings only and treating the task as binary acoustic anomaly detection. The data contains multiple machine cases and recording channels. Although the provided anomaly-condition metadata describes combinations of conditions rather than clean single-cause labels, we do not evaluate root-cause classification in this paper.

We compare the training-free prompting approach against a one-class SVM baseline. The one-class SVM requires fitting a task-specific one-class decision function from normal data, while the LLM setting uses only the matched reference clips provided in the prompt. Main results are reported on a held-out test split using *Accuracy* and *F1*, with the primary comparison focused on the 4-shot and 16-shot regimes.

3.2 Main Results and Discussion

Table 1 shows a clear crossover between the proposed prompting approach and the one-class baseline. At 4-shot, GPT-5.4-mini outperforms one-class SVM on the held-out test split in both *Accuracy* and *F1*. This suggests that, when only a very small number of matched normal examples are available, structured LLM prompting can act as a useful training-free comparator.

This advantage does not persist as more matched normal references are added. At 16-shot, one-class SVM becomes clearly stronger than all tested LLM settings and achieves the best overall performance. With more normal data, a fitted one-class detector can learn a more stable task-specific boundary, whereas the LLM still operates only through prompt-level comparison.

These results indicate that the main value of the proposed method is not universal superiority over classical anomaly detectors, but low-shot flexibility. In the smallest-reference regime, structured prompting over compact acoustic summaries can be competitive with, and sometimes stronger than, a classical baseline. As the number of normal references increases, the practical case shifts toward conventional normal-only training.

Pure zero-shot prompting was unreliable, and ultra-low-shot settings were unstable, indicating that some matched normal context is necessary. Performance improved from 4-shot to 8-shot, but showed little further gain beyond 16-shot. Overall, acoustic anomaly detection appears better framed as a reference-based comparison than as an absolute zero-shot judgment, with the practical benefit of LLM prompting concentrated in a narrow low-shot regime.

4 CONCLUSION

We presented an early exploration of reference-based LLM prompting for acoustic anomaly detection. The key result is not that LLMs replace classical detectors, but that they provide a viable training-free comparison mechanism when only a few matched normal examples are available. Our experiments show that zero-shot prompting is unreliable, matched normal references are important, and the advantage of prompting is strongest in the low-shot regime. Future work should study stronger non-LLM few-shot baselines, lightweight local language models, and broader evaluation across machine types and recording conditions.

ACKNOWLEDGMENTS

This work was partly supported by the National Edge AI Hub for Real Data: Edge Intelligence for Cyberdisturbances and Data Quality (UKRI EPSRC EP/Y028813/1).

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